

Multiple Choice (20 marks)

1. Which of the following is an example of genetic engineering?
 - (a) Growing a whole plant from a single cell.
 - (b) Finding the sequences of bases in plant DNA.
 - (c) Inserting a gene into plants that makes them resistant to insects.
 - (d) Attaching the root of one type of plant to the stem of another type of plant.
2. Hume argues that the criteria for judging aesthetic value are _____.
 - (a) entirely subjective
 - (b) unchangeable
 - (c) entirely objective
 - (d) unknown
3. SWOT is an acronym for:
 - (a) Strengths, weaknesses, opportunities, threats.
 - (b) Strategy, work, openness, toughness.
 - (c) Strategy, weakness, opinions, tactics.
 - (d) Strategy, working, opinion, tactical.
4. A person is 6 feet tall. Which of the following metric measurements is most near to 6 feet?
 - (a) 6 meters
 - (b) 3 meters
 - (c) 2 meters
 - (d) 1 meter
5. Dopamine is prescribed at a rate of 4 micrograms/kg/min. For a 65 kg person, how many milligrams will they receive in an hour?
 - (a) 156
 - (b) 15.6
 - (c) 1.56
 - (d) 15600
6. Arguing that someone should do something because other people are doing it is a description of which of the following fallacies?

- (a) appeal to common practice
 - (b) snob appeal
 - (c) argument by consensus
 - (d) appeal to tradition
7. Which features are physical properties of an iron nail?
- (a) conducts heat, magnetic, solid
 - (b) conducts electricity, magnetic, rusts
 - (c) conducts sound, insulates, non-magnetic
 - (d) dissolves, conducts heat, non-magnetic
8. A student uses clay to build models of an ocean plate and a continental plate. What characteristic cannot be accurately represented by the models?
- (a) the shapes of plates
 - (b) the relative sizes of plates
 - (c) the relative densities of plates
 - (d) the arrangement of plates relative to one another
9. If people expect the price of a particular product to increase in the near future
- (a) this will not affect the demand for the product right now.
 - (b) this will decrease the demand for the product.
 - (c) this will not affect the demand for the product now or later.
 - (d) this will increase the demand for the product.
10. In the assessment of the hand function which of the following is true?
- (a) Abduction of the thumb is supplied by spinal root T2
 - (b) Opposition of the thumb by opponens pollicis is supplied by spinal root T1
 - (c) Finger adduction is supplied by the median nerve
 - (d) Finger abduction is mediated by the palmar interossei

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to round the given number to the nearest multiple of a specific number.
2. Write a python function to find the cube sum of first n even natural numbers.

End of Paper